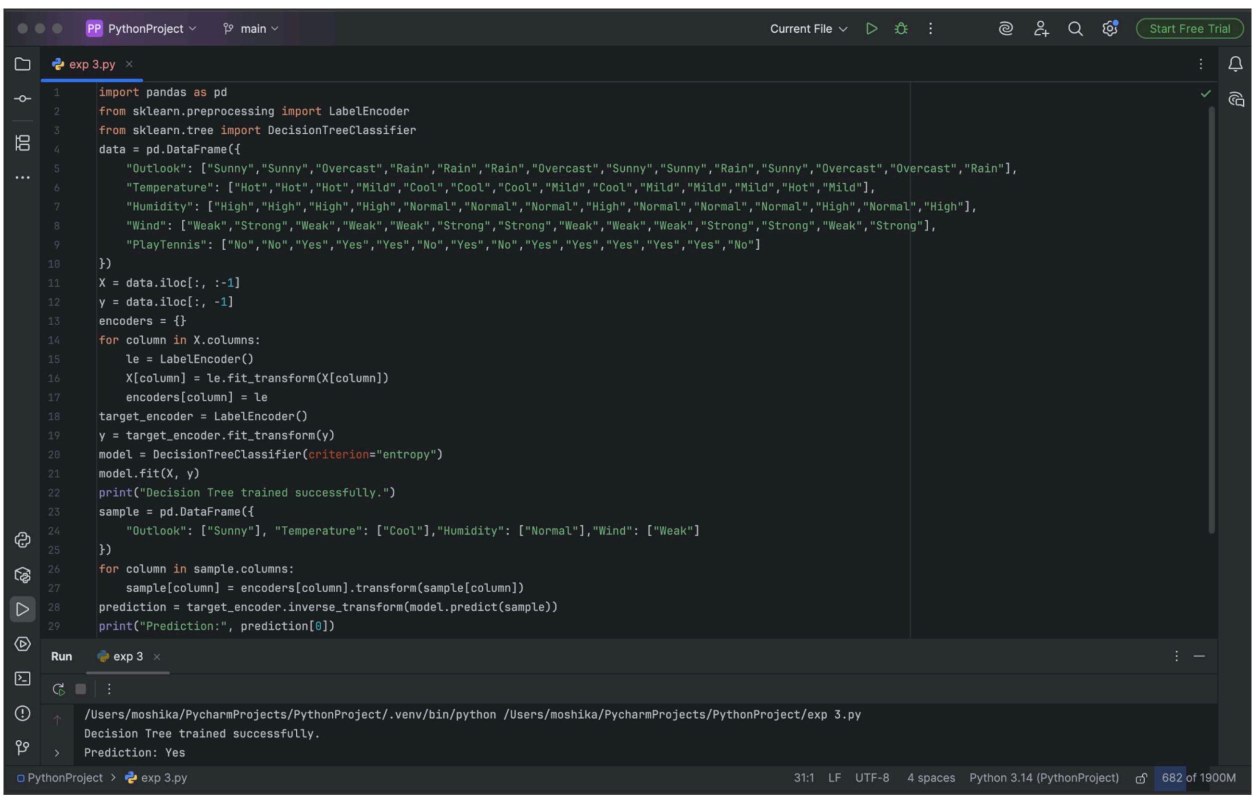
EXPERIMENT - 3

Aim: To construct a decision tree using the ID3 algorithm and classify a new sample.

Procedure:

1. Load the dataset.
2. Calculate the entropy of the target attribute.
3. Calculate information gain for each attribute.
4. Select the attribute with the highest information gain.
5. Create a decision tree node using that attribute.
6. Divide the dataset based on its attribute values.
7. Repeat the process recursively for each subset.
8. Stop when the subset belongs to one class.
9. Use the generated tree to classify a new sample.

Output:



```python
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
data = pd.DataFrame({
    "Outlook": ["Sunny","Sunny","Overcast","Rain","Rain","Rain","Overcast","Sunny","Sunny","Rain","Sunny","Overcast","Overcast","Rain"],
    "Temperature": ["Hot","Hot","Hot","Mild","Cool","Cool","Cool","Mild","Cool","Mild","Mild","Mild","Hot","Mild"],
    "Humidity": ["High","High","High","High","Normal","Normal","Normal","High","Normal","Normal","Normal","High","Normal","High"],
    "Wind": ["Weak","Strong","Weak","Weak","Weak","Strong","Strong","Weak","Weak","Weak","Strong","Strong","Weak","Strong"],
    "PlayTennis": ["No","No","Yes","Yes","Yes","No","Yes","No","Yes","Yes","Yes","Yes","Yes","No"]
})
X = data.iloc[:, :-1]
y = data.iloc[:, -1]
encoders = {}
for column in X.columns:
    le = LabelEncoder()
    X[column] = le.fit_transform(X[column])
    encoders[column] = le
target_encoder = LabelEncoder()
y = target_encoder.fit_transform(y)
model = DecisionTreeClassifier(criterion="entropy")
model.fit(X, y)
print("Decision Tree trained successfully.")
sample = pd.DataFrame({
    "Outlook": ["Sunny"], "Temperature": ["Cool"],"Humidity": ["Normal"],"Wind": ["Weak"]
})
for column in sample.columns:
    sample[column] = encoders[column].transform(sample[column])
prediction = target_encoder.inverse_transform(model.predict(sample))
print("Prediction:", prediction[0])
```

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 3.py
Decision Tree trained successfully.
Prediction: Yes
```

Result: The ID3 algorithm successfully constructs a decision tree and correctly classifies the given new sample.